

GREATER ROCHESTER INTL, NY, USA

WMO: 725290

Lat: 43.117N

Lon: 77.677W

Elev: 539

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14768

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 6.6	(d) -5.1	(e) 4.3	(f) 3.8	(g) -1.2	(h) 5.3	(i) 8.6	(j) 30.5	(k) 32.8	(l) 27.3	(m) 29.5	(n) 9.4	(o) 240	(p) 0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.7	(c) 88.9	(d) 73.0	(e) 86.0	(f) 71.4	(g) 83.2	(h) 69.8	(i) 75.4	(j) 84.5	(k) 73.7	(l) 82.1	(m) 72.2	(n) 80.1	(o) 11.6	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	123.0	80.4	70.9	116.3	78.3	69.3	110.0	76.9	39.2	84.7	37.5	82.4	36.2	80.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature										n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years									
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max								
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)										
25.2	21.5	19.0	-3.6	92.5	6.3	2.7	-8.1	94.5	-11.8	96.0	-15.3	97.6	-19.9	99.5										
			-4.1	78.2	6.1	2.0	-8.5	79.6	-12.1	80.8	-15.5	81.9	-19.9	83.4										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	25.4	27.0	34.7	46.3	58.1	67.1	71.5	70.0	63.4	52.0	41.0	31.7
	DBStd	18.48	11.75	10.50	10.68	9.17	8.47	6.72	5.52	5.34	7.24	8.31	8.95	9.22
	HDD50	3041	766	646	490	178	19	0	0	0	1	75	295	571
	HDD65	6404	1228	1064	940	566	249	54	9	15	115	410	720	1034
	CDD50	2729	3	2	16	66	271	514	668	621	402	138	25	3
	CDD65	615	0	0	0	4	36	119	211	171	66	8	0	0
	CDH74	4993	0	0	7	58	360	998	1768	1244	501	57	0	0
	CDH80	1413	0	0	0	10	90	291	551	333	132	6	0	0
Wind	WSAvg	9.0	10.8	10.6	10.3	10.1	8.7	7.9	7.4	6.9	7.3	8.6	9.3	9.9
Precipitation	PrecAvg	31.5	2.1	1.8	2.2	2.6	2.7	3.1	2.8	3.5	3.1	2.4	2.9	2.3
	PrecMax	45.1	5.6	3.9	4.7	4.8	6.2	7.1	6.3	6.0	6.3	5.7	7.0	4.2
	PrecMin	21.0	0.6	0.6	1.0	1.2	0.4	0.2	0.6	0.8	0.3	0.2	0.4	0.8
	PrecStd	5.2	1.1	0.8	0.9	1.0	1.4	1.7	1.5	1.3	1.4	1.2	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	61.1	72.2	81.1	87.2	90.5	92.2	91.4	89.4	80.1	68.6	60.9
		MCWB	54.7	52.1	59.3	63.0	69.9	73.7	74.8	74.1	73.1	66.6	57.8	53.4
	2%	DB	53.5	52.9	62.8	73.6	82.3	86.8	89.0	86.9	83.6	74.5	64.1	53.7
		MCWB	49.6	45.9	52.8	58.5	66.9	71.3	73.4	72.5	70.7	63.6	55.5	48.1
	5%	DB	46.8	46.6	57.0	67.4	78.0	83.4	85.8	83.5	79.6	69.9	60.1	49.5
		MCWB	42.8	41.0	49.4	55.1	64.7	69.8	71.6	70.5	68.4	61.4	53.1	45.2
	10%	DB	41.2	41.0	50.9	62.1	73.4	79.9	82.8	80.7	75.9	65.3	55.0	44.5
		MCWB	37.6	36.9	43.8	52.0	61.8	67.8	69.9	69.2	66.4	57.9	48.9	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	53.9	60.4	64.8	72.5	76.6	78.3	77.4	75.0	69.7	60.1	55.1
		MCDB	60.7	59.4	70.6	77.5	83.8	86.3	88.3	87.7	84.6	76.3	65.6	59.9
	2%	WB	49.6	46.8	54.6	60.4	69.2	73.8	75.7	74.7	72.6	65.5	57.1	49.4
		MCDB	53.0	52.3	61.1	70.2	78.9	83.0	85.0	82.9	81.0	72.1	62.5	52.5
	5%	WB	43.3	41.4	49.8	57.0	66.6	71.8	73.8	72.9	70.1	62.1	53.6	44.9
		MCDB	46.6	46.4	56.5	65.9	75.5	80.2	82.2	80.4	77.1	68.5	59.3	49.2
	10%	WB	37.7	36.8	44.2	52.8	63.8	69.9	72.0	71.1	67.9	59.3	49.8	40.6
		MCDB	40.8	40.6	50.2	60.9	71.7	77.2	79.9	78.0	74.4	64.8	54.7	44.2
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.5	15.6	18.5	19.7	18.8	18.7	18.2	18.6	16.4	14.2	11.8
		MCDBR	18.3	21.9	24.6	28.2	24.9	22.7	21.7	21.6	22.6	22.7	21.6	17.6
		MCWBR	15.9	17.2	16.6	16.6	13.0	11.0	9.9	10.1	11.4	13.7	15.2	14.6
	5% WB	MCDBR	17.7	20.7	23.2	25.2	22.1	19.6	18.4	18.2	19.6	20.3	19.1	16.6
		MCWBR	16.1	17.2	16.6	16.7	12.9	11.2	9.8	9.5	10.6	13.4	15.1	14.8
Clear-Sky Solar Irradiance	taub	0.306	0.316	0.335	0.371	0.395	0.436	0.433	0.421	0.374	0.357	0.313	0.294	
	taud	2.380	2.355	2.389	2.339	2.320	2.235	2.255	2.274	2.391	2.427	2.511	2.483	
	Ebn at Noon	262	279	287	284	279	266	266	265	271	261	259	258	
	Edh at Noon	25	30	33	38	40	43	42	40	33	28	22	21	
All-Sky Solar Radiation	RadAvg	411	663	1059	1421	1735	1844	1869	1663	1325	798	494	327	
	RadStd	38	66	102	152	176	151	144	94	114	75	66	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	N/A

Nomenclature: See separate page